Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that port 53 is unreachable whenever attempts are made to access the client company website, www.yummyrecipesforme.com. Port 53 is used for Translating domain names to IP addresses. The DNS server is not responding, so this means that UDP port 53 is not available on the server

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
The incident occurred around noon when several customers of clients reported that they were not able to access the client company's website. Myself and other analysts responded. We visited the site, but received the error, 'destination port unreachable'. We began to troubleshoot the issue by loading the network analyzer tool tcpdump and tried loading the page again. By loading the page again, a query was sent to a DNS server via the UDP protocol through the browser in order to receive the IP address for the website's domain name. After which, the browser uses the IP address as the destination IP for sending an HTTPS request to the web server in order to properly display the webpage. But, the result for that was a ICMP packet that said, 'udp port 53 is unreachable.' The most likely causes of this incident is that the DNS IP address given was wrong, or the DNS server is not listening in to udp port 53, the DNS server has crashed or there's already a firewall put in place that is rejecting traffic to udp port 53. Moving forward, we verify the accuracy of the DNS IP address, we check for possible firewalls, ensure that the DNS server is listening to udp port 53 and also ensure that the DNS server is secure and stable, or we try using a relatable DNS server if the current one has unresolved issues